

Qns	Answer	Marks
5(a)(i)	At 22.5 °C, $R_T = 1.6 \text{ k}\Omega$ (read from graph)	1
	Total resistance = $\left(\frac{1}{1600} + \frac{1}{1600}\right)^{-1} = 800 \text{ }\Omega$	1
(a)(ii)	Using potential divider principle, $\frac{V}{9.0} = \frac{800}{800 + 1200}$ $V = 3.6 \text{ V}$	1
		1
(b)(i)	Using potential divider principle, $\frac{1200}{R_{AB}} = \frac{5}{4}$ $R_{AB} = 960 \text{ }\Omega$	1
		1
(b)(ii)	$\frac{1}{960} = \frac{1}{1600} + \frac{1}{R_T}$ $R_T = 2400 \text{ }\Omega$	1
	From graph, temperature = 10.5 °C OR 10.75 °C (read to half the smallest division) OR 11 °C	1
(c)	From graph, <u>no calibration data available between 24°C to 25°C (accept "at 25 °C"O).</u>	1
	Since graph is <u>non-linear</u> , hence <u>cannot extrapolate</u> to get values of temperature. Based on temperature range of 0 – 25 °C, V_{AB} has a range of 0.48 V to 3.5 V. Since only a <u>small part of the voltage scale</u> is used, the <u>measurement of temperature will be less sensitive</u> .	1

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6(e)(i)	Progressive wave is one where energy is transferred in the direction of	1